

Question 5

Take a number, repeatedly sum its digits until the result is a single digit, and display each intermediate result. For example, if the number is 123, the sum of its digits is 6.

Pseudocode

Begin

Declare N and Sum as integers

Input N

original_num $\leftarrow$ N

while $N \geq 10$

Sum $\leftarrow$ 0

while $N > 0$

Digit $\leftarrow$ $N \% 10$

Sum $\leftarrow$ Sum + Digit

$N \leftarrow$ N Divided by 10

end while

$N \leftarrow$ Sum

Output N

end while

if original_num < 10

Output original_num

end if

end

Algorithm

1. start
2. Input number N
3. Repeat while N has two or more digits.
 - Set $sum = 0$
 - repeat while $N > 0$
 - Find the last digit using $N \text{ Mod } 10$
 - Add the digit to sum
 - Remove the last digit using $N/10$
 - Set $N = \text{Sum}$
 - Display Sum
4. If original number was already a single digit, display original number.
5. Stop